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Mehedi Hasan

Nationality: Bangladeshi **Date of birth:** 3 Mar 1997

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Other: Theodebertstrasse-11, top 6, 5020 Salzburg (Austria)

ABOUT ME

Geography enthusiast specializing in **multi-hazard risk assessment, vulnerability analysis, and disaster risk reduction**, with expertise in GIS, Python, remote sensing, and machine learning to support data-driven disaster management and resilient communities.

EDUCATION AND TRAINING

Master of Science (M.Sc.), Professional

University of Salzburg [5 Oct 2026 – Current]

City: Salzburg | **Country:** Austria | **Website:** https://www.plus.ac.at/geoinformatik/ | **Field(s) of study:** Applied Geoinformatics | **Level in EQF:** EQF level 7

Master of Science (M.Sc.), General

Jagannath University [5 Feb 2020 – 16 Mar 2021]

City: Dhaka | **Country:** Bangladesh | **Website:** https://jnu.ac.bd/department/portal/geography_environment | **Field(s) of study:** Geography and Environment | **Final grade:** 3.43 out of 4 | **Level in EQF:** EQF level 7 | **Thesis:** Assessment of Moderate Seismic Vulnerability and Earthquake Preparedness in Rangpur Division, Bangladesh

Bachelor of Science (B.Sc.)

Jagannath University [5 Feb 2015 – 16 Sep 2019]

City: Dhaka | **Country:** Bangladesh | **Website:** https://jnu.ac.bd/department/portal/geography_environment | **Field(s) of study:** Geography and Environment | **Final grade:** 3.27/4(CGPA) | **Level in EQF:** EQF level 6

Project Title: Urban Fire Hazard Risk and Preparedness in Crowded Settlements: The Case of Sadarghat, Dhaka, Bangladesh.

RESEARCH INTERESTS

Flood Risk | Disaster Resilience | GIS & Remote Sensing | Geospatial Analysis | Machine Learning (ML) | Deep Learning (DL) | Hydrological Modeling | Climate Change Impact | Risk Assessment | Urban & Regional Planning

PUBLICATIONS

[2026]

Multi-hazard risk mapping using geospatial techniques in coastal Bangladesh: implications for resilience

Authors: Sharmin Akter, Muhammad Al-Amin Hoque, Mehedi Hasan, and Biswajeet Pradhan | **Journal Name**: Geomatics, Natural Hazards and Risk | **Volume, Issue and Pages**: 17(1) | **Publisher**: Taylor & Francis

Link: https://doi.org/10.1080/19475705.2026.2729590

[2026]

Machine learning-driven flood susceptibility mapping in the coastal frontier of Feni District, Bangladesh

Authors: Maliha Sanzana Misty, Mehedi Hasan, Muhammad Al-Amin Hoque, Md. Kamrul Islam, A. M. M. Jubayer Nayem, and Nazim Uddin. | **Journal Name**: Geomatics, Natural Hazards and Risk | **Volume, Issue and Pages**: 17(1) | **Publisher**: Taylor & Francis

Link: https://doi.org/10.1080/19475705.2026.2717863

[2025]

Wetland change and its impacts on livelihood using machine learning algorithms in Chanda Beel, Gopalganj, Bangladesh

Authors: Nazim Uddin, Mehedi Hasan, Md. Nadim Sheikh, and Asib Ahmed | **Journal Name**: Next Research | **Volume, Issue and Pages**: 8,1-13 | **Publisher**: Elsevier

Link: 10.1016/j.nexres.2026.101556

[2025]

A Comparative Study of Spatial MCDM Techniques for Flash Flood Risk Mapping in Northeastern Bangladesh

Journal Name: Earth Systems and Environment | **Volume, Issue and Pages**: 10, 6077–6095 | **Publisher**: Springer Nature

Muhammad Al-Amin Hoque, Md. Shamsudduha Sami, Mehedi Hasan, and Biswajeet Pradhan

Link: https://link.springer.com/article/10.1007/s41748-025-00942-2

[2025]

Mapping seismic risk in Rangpur Division, Bangladesh: An integrated geospatial approach.

Authors: Mehedi Hasan, Muhammad Al-Amin Hoque, and Biswajeet Pradhan. | **Journal Name**: Spatial Information Research | **Volume, Issue and Pages**: 33(1), 40 | **Publisher**: Springer Nature

Link: https://link.springer.com/article/10.1007/s41324-025-00643-y

[2026]

Flood susceptibility prediction using machine learning under hydro-climatic extremes and land use dynamics in north western region, Bangladesh

Authors: Hasan, M., Uddin, N., Misty, M. S., & Hoque, M. A | **Journal Name**: Journal of Environmental Management | **Volume, Issue and Pages**: Manuscript ID: JEMA-D-26-17551 | **Publisher**: Elsevier

[2026]

A novel ensemble machine learning framework to identify groundwater risk zone for sustainable water resource planning in Jashore District, Bangladesh.

Authors: Roy, S. K., Dey, S., Mahmudul Hasan, M., Hasan, M., Majumder, P., & Porun Rana, M. M. | **Journal Name**: Journal of Hydrology and Earth System Sciences | **Volume, Issue and Pages**: Manuscript ID: JHESS-D-26-04567 | **Publisher**: Elsevier

CONFERENCES AND SEMINARS

[18 Dec 2025 – 19 Dec 2025] Jagannath University, Dhaka, Bangladesh

International Conference on Life, Earth and Science for Sustainable Development (ICLESSD-2025) Presented my research, “Predicting Thermal Comfort Impacts of Urban Water Shrinkage in Dhaka Using Artificial Learning Models,” at ICLESSD-2025. The study used time-series analysis to examine changes in urban water bodies and temperature in Dhaka and investigate their relationship with urban thermal comfort using machine learning models.

[21 Nov 2025 – 22 Nov 2025] Asian Institute of Technology (AIT), Bangkok, Thailand

International Workshop on Technologies for Adaptation to Climate Change (IWTACC-2025) Presented the research paper "Assessing Climate Change Adaptive Capacity using Geospatial Techniques: Application to a Coastal Upazila in Bangladesh" at the International Workshop on Technologies for Adaptation to Climate Change (TACC 2025). The study used geospatial techniques to assess climate change adaptive capacity and identify spatial variations in the adaptive capacity of coastal communities in Bangladesh.

WORK EXPERIENCE

Institute of Water Modelling (IWM) – Dhaka, Bangladesh

Website: https://iwmbd.org | **Name of unit or department:** ICT, GIS & Remote Sensing Division (ICT) - **Business or sector:** Administrative and support service activities

Link https://iwmbd.org/Divisions/ICT

GIS Analyst

[1 Apr 2024 – 1 Sep 2026]

OVIBASHI KARMI UNNAYAN PROGRAM (OKUP) – Dhaka, Bangladesh

Website: https://www.okup.org.bd

Link https://www.okup.org.bd

Research Assistant

[1 Feb 2023 – 10 Mar 2024]

Geomark Limited – Dhaka, Bangladesh

Website: https://geomark.com.bd | **Business or sector:** Administrative and support service activities

GIS Operator

[3 Aug 2022 – 31 Oct 2022]

- Digitization and geospatial data quality checking
- GIS mapping and spatial analysis for environmental and urban planning projects
- Handling and processing large geospatial datasets

Dr. Muhammad Al-Amin Hoque (Professor) – Dhaka, Bangladesh

Website: https://jnu.ac.bd | **Email address:** hoque@geography.jnu.ac.bd | **Name of unit or department:** Department of Geography and Environment

Link https://jnu.ac.bd/profile/portal/web/549

Research Assistant

[10 Mar 2021 – 29 Dec 2022]

PRIME ACADEMY AND CONSULTANCY – Dhaka, Bangladesh

Address: 13/1C, Bhojohari Street, Wari, 1203 Dhaka (Bangladesh)

Assistant Teacher (Science)

[Apr 2019 – Mar 2020]

CERTIFICATIONS

[KIM's GIS, 5 Jun 2025]

Practical GIS & Remote Sensing with Python and ArcPy: Step-by-Step Learning

Mode of learning: Online

[National Academy for Computer Training and Research (NACTAR), 1 Jan 2025]
Machine Learning and Data Science

Mode of learning: Online

Link: https://nactar.gov.bd

[Youth Development Program , 21 Jul 2022]
Computer Training Program

Mode of learning: Presential

Link: https://dyd.gov.bd/

[Institute of GIS, AI & RS, 30 Jun 2022]
Google Earth Engine (GEE)

Mode of learning: Online

Link: https://www.studyhacksgeospatial.com/

[Group Mappers, 10 Feb 2022]
Application of GIS

Mode of learning: Presential

Link: https://groupmappers.com

SKILLS

Hard Skills

Python Programming / Machine Learning(ML) / Deep Learning(DL) / JAVA (for GEE) / R(Basic) / SQL / Spatial Data Analysis / GIS & Remote Sensing / AHP & MCDA / Flood & Cyclone Risk Assessment / GIS& RS

Software and Tools

Anaconda / Power BI / PyCharm / TerrSet / Jupyter lab / HEC-RAS / ArcGIS/Pro / QGIS / Microsoft Office / Adobe Photoshop / MATLAB / Seismograph / MIKE SHE

HONOURS AND AWARDS

[2017] Dept. of Geography and Environment, JnU
Best Presenter Award

[2018] Dept. of Geography and Environment, JnU
Champion & Best Defender Award

[2018] Desh Sangbad
Best Content Award

[2019] Youth Journalist Forum, JnU
Publication Secretary

[2018] Environmental Club, JnU
Member

[2021] Green Voice, Dhaka.
Member

LANGUAGE SKILLS

Mother tongue(s): Bangla

Other language(s):

English

LISTENING C1 **READING** B2 **WRITING** B2

SPOKEN PRODUCTION C1 **SPOKEN INTERACTION** C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

RECOMMENDATIONS

Name: **Dr. Muhammad Al-amin Hoque** | Professor

Geography and Environment Department Jagannath University, Bangladesh.
Email: hoque@geography.jnu.ac.bd | **Phone number:** (+880) 01797599361

Name: **Md. Zahid Hasan Siddiquee** | Associate GIS/RS Specialist
ICT, GIS & Remote Sensing Division (ICT)
Institute of Water Modelling (IWM)
Uttara, Bangladesh.
Email: zhs@iwmbd.org | **Phone number:** (+880) 01714072557